

VITEBSK, Belarus

WMO: 266660

Lat: 55.127N

Lon: 30.350E

Elev: 682

StdP: 14.34

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.8	-0.8	-12.2	2.9	-6.2	-6.3	4.0	0.6	21.6	30.0	19.0	27.5	3.7	20	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	83.9	68.0	80.5	66.2	77.3	64.5	69.6	80.6	67.8	77.4	66.2	75.0	6.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	97.6	74.6	64.3	92.4	72.4	62.7	87.2	71.1	34.0	81.1	32.5	77.7	31.3	75.1	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.2	14.3	-14.2	88.5	7.0	4.0	-19.2	91.4	-23.3	93.7	-27.2	96.0	-32.3	98.9		
(5)			-14.6	72.4	6.9	1.6	-19.5	73.6	-23.6	74.5	-27.5	75.4	-32.5	76.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.0	22.0	22.8	31.6	45.1	55.6	62.0	66.5	63.8	54.0	43.2	33.5	25.7		
	DBStd	17.86	11.16	11.49	8.20	8.26	8.26	6.15	5.57	6.24	6.34	7.29	8.62	10.45		
	HDD50	3943	869	761	570	192	38	1	0	0	33	233	494	752		
	HDD65	7896	1334	1181	1035	596	308	129	47	97	331	677	944	1217		
	CDD50	1735	0	0	1	46	212	361	512	428	154	21	0	0		
	CDD65	213	0	0	0	0	17	40	94	60	2	0	0	0		
	CDH74	1547	0	0	0	4	134	225	669	493	22	0	0	0		
	CDH80	349	0	0	0	0	21	25	157	145	1	0	0	0		
	WSAvg	6.6	7.0	6.8	7.2	6.9	6.6	6.3	5.7	5.9	6.0	6.7	7.1	7.3		
	PrecAvg	28.9	2.2	1.9	1.7	1.5	2.5	3.1	3.5	3.0	2.6	2.6	2.3	2.0		
	PrecMax	36.6	4.2	3.3	3.9	3.4	4.5	5.8	8.2	11.4	5.5	6.0	4.5	3.5		
(15) Precipitation	PrecMin	21.8	0.4	0.7	0.5	0.4	0.7	0.8	0.8	0.3	0.7	0.5	0.0	0.8		
	PrecStd	4.0	0.9	0.7	0.7	0.8	1.1	1.3	1.9	2.2	1.4	1.4	1.0	0.7		
	DB	42.0	43.0	56.9	73.3	82.7	83.6	88.2	91.3	77.0	66.1	52.9	46.6			
	MCWB	40.3	39.7	45.8	55.8	65.6	67.6	69.2	69.4	63.1	55.6	50.3	45.0			
	2% DB	38.4	39.0	50.3	67.9	77.7	79.8	84.4	83.1	71.5	59.0	49.3	43.2			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	36.9	36.5	41.4	52.8	63.4	65.2	68.7	67.2	60.1	53.6	47.3	41.9			
	5% DB	36.2	37.0	45.8	62.9	73.4	76.7	80.8	79.1	67.4	55.8	46.3	40.1			
	MCWB	34.9	35.0	39.5	50.5	60.2	64.0	67.2	65.4	58.4	52.0	44.3	39.0			
	10% DB	34.2	35.3	42.0	58.9	69.5	73.5	77.4	75.3	64.2	53.3	44.0	37.0			
	MCWB	33.1	33.8	37.6	48.8	57.7	62.3	65.4	63.9	56.6	50.2	42.5	35.8			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	40.7	40.3	46.8	57.5	67.8	69.8	72.6	71.0	64.7	57.4	51.1	45.3			
	MCDB	41.4	42.4	54.6	69.8	79.2	79.3	83.8	85.6	74.4	62.3	52.5	46.0			
	2% WB	36.9	37.0	43.2	54.2	64.8	67.5	70.4	68.9	61.4	55.0	47.6	42.0			
	MCDB	37.8	38.3	48.7	63.7	75.2	76.0	80.9	80.4	67.9	58.1	48.9	42.8			
	5% WB	35.0	35.3	40.3	52.1	61.9	65.7	68.3	67.1	59.4	52.6	44.6	38.9			
(31) Mean Daily Temperature Range	MCDB	35.9	36.5	44.0	60.2	71.4	73.8	78.1	76.6	65.2	55.1	45.7	40.0			
	10% WB	33.3	33.9	38.0	49.8	59.3	63.7	66.6	65.1	57.6	50.4	42.6	35.8			
	MCDB	34.0	34.9	41.5	57.8	67.3	71.2	75.5	73.1	62.7	53.1	43.8	36.8			
	5% DB	8.0	10.0	11.8	15.4	16.9	15.6	16.3	16.7	14.7	10.3	7.0	6.9			
	MCDBR	6.3	7.6	16.2	21.7	21.6	19.7	20.5	22.5	20.1	13.7	8.9	6.5			
(36) Clear-Sky Solar Irradiance	MCWBR	6.2	6.2	10.1	10.6	10.2	9.1	8.8	9.5	10.7	9.4	7.4	6.0			
	5% WB	5.8	6.6	13.7	18.9	20.1	17.0	18.9	19.2	17.6	11.7	7.8	6.4			
	MCWBR	5.9	5.8	9.4	10.3	10.1	9.1	9.0	9.0	10.5	8.8	7.1	6.2			
	taub	0.287	0.320	0.348	0.401	0.390	0.374	0.401	0.410	0.367	0.347	0.327	0.289			
	taud	2.154	2.108	2.156	2.202	2.312	2.379	2.320	2.287	2.378	2.394	2.253	2.172			
(42) All-Sky Solar Radiation	Ebn at Noon	197	232	257	259	270	276	266	254	251	226	181	167			
	Edh at Noon	20	30	36	40	38	36	37	36	29	23	18	16			
	RadAvg	194	476	921	1267	1644	1778	1712	1379	907	429	154	116			
(45)	RadStd	45	78	125	151	151	163	160	141	75	88	28	32			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (7 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page